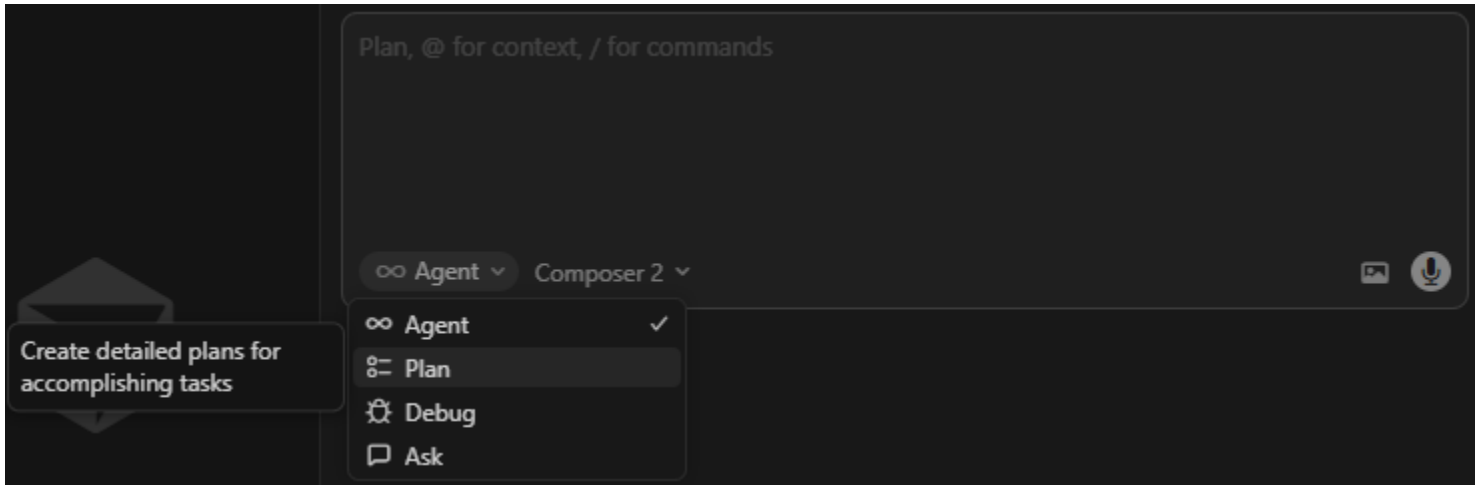
Second Exercise: AI-Assisted Development

- We tried vibe coding. Now let’s take back control.
- In this exercise, we will be using Cursor to help us code again, but the project will be larger and we will try three different techniques: **Planning, Incremental Improvement,** and **Managing**
- The project these slides are working on is: *building an interactive call graph visualizer for x86 binaries*
- You can follow along with that goal, or create your own if you wish

Second Exercise: AI-Assisted Development

- Technique 1: Planning
  - Planning helps the agent do more complicated tasks, with a more well-specified goal for each step. It also allows you to see what design decisions the AI is making and approve or correct them as needed.
  - In the bottom left of the chat window, you can change your chat session type to “Agent” or “Plan” or “Ask”



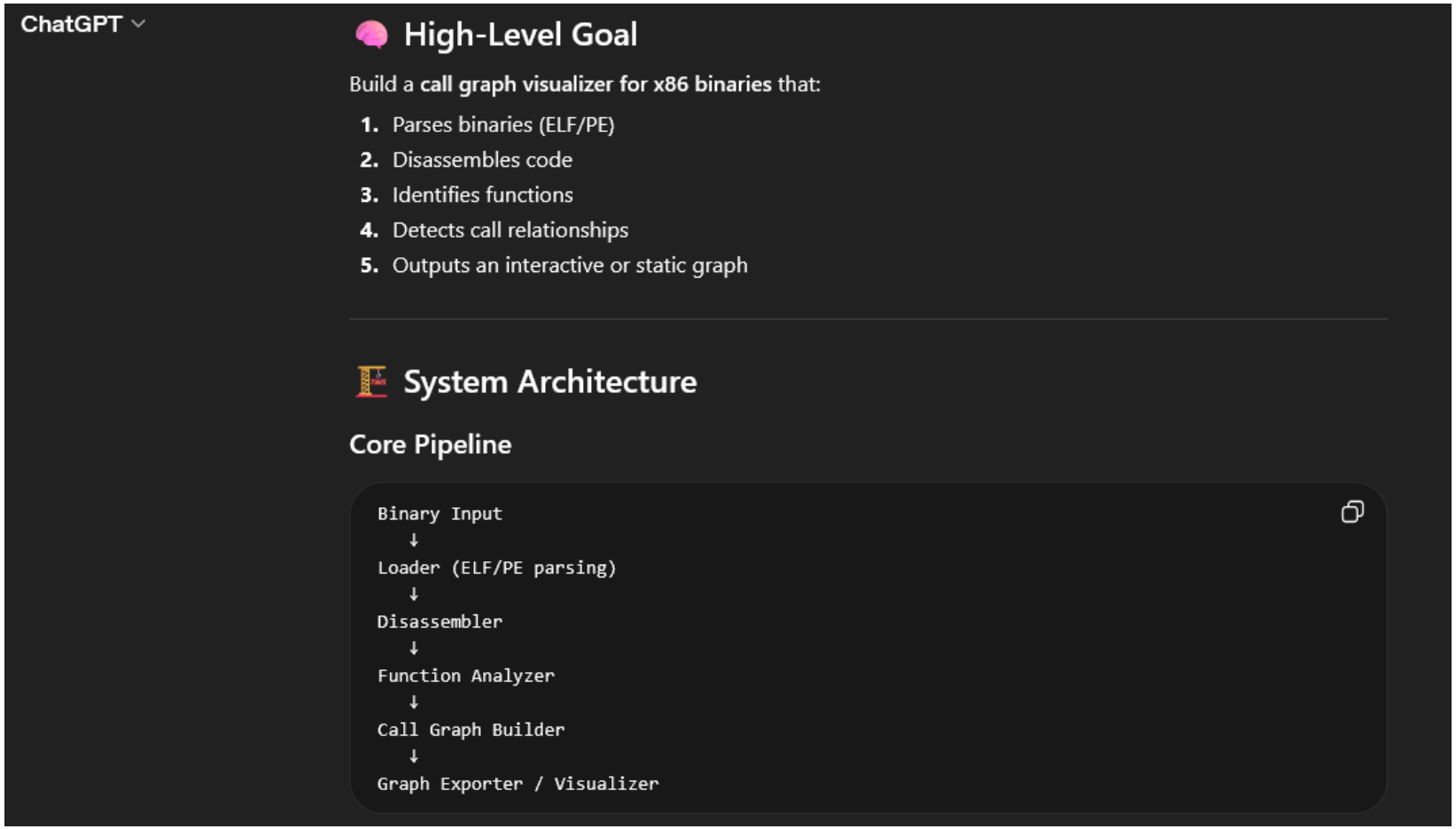
Second Exercise: AI-Assisted Development

- Technique 1: Planning
  - **Agent:** this mode is the default that we used earlier, it can do basic coding tasks.
  - **Ask:** this mode is for asking questions when you DON’T want the AI to write code. You can ask about the code base, ask it for ideas, etc.
  - **Plan:** this mode is for larger, more complicated coding tasks. Before writing any code, this mode will create a plan document that is essentially a series of prompts that the agent will use to break down and accomplish larger coding tasks.

Second Exercise: AI-Assisted Development

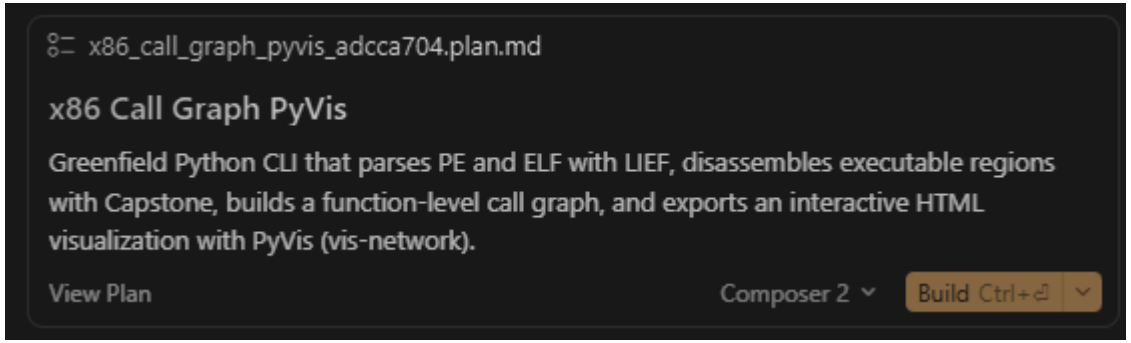
- Technique 1: Planning
  - *(Optional)* You can also try co-designing your software's MVP with a different AI, like through ChatGPT. Almost any AI can be helpful to help synthesize a design document and help find holes and problems and solutions.
  - **MVP: Minimum Viable Product.** What is the smallest subset of your intended program that you can build to have something that runs and can be tested/used?

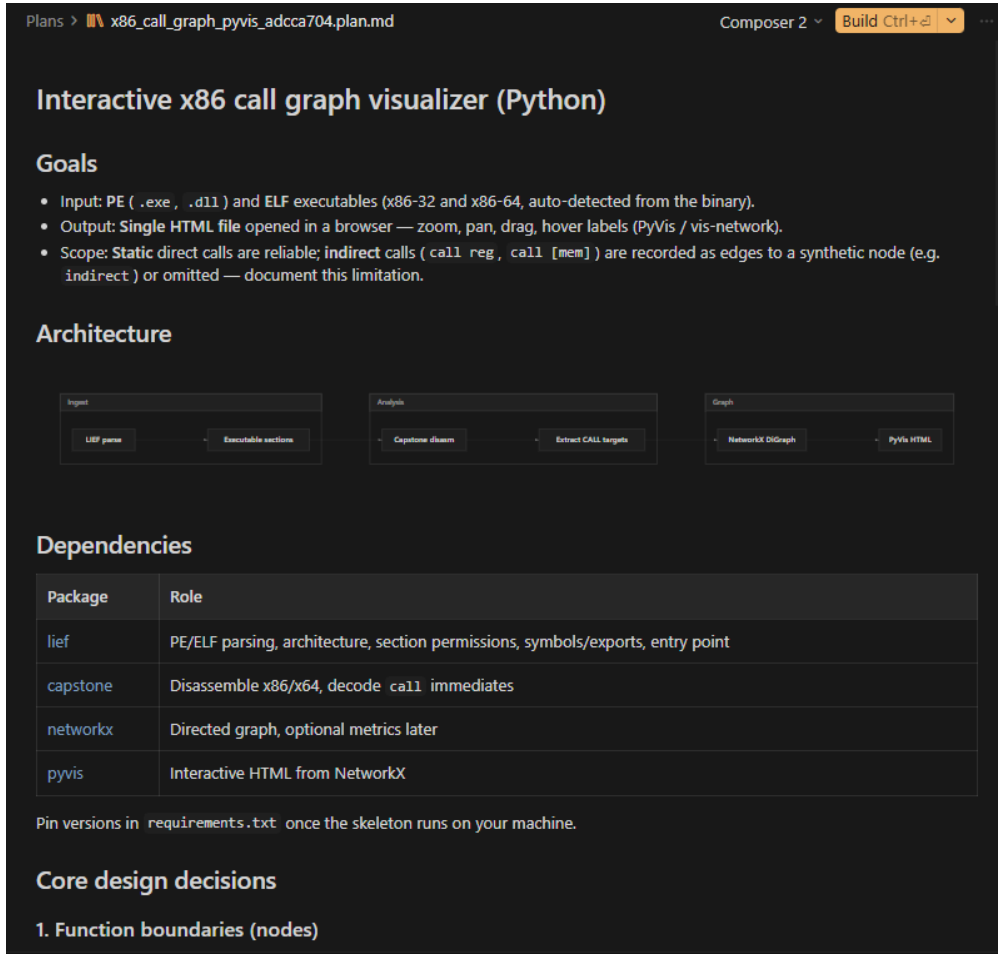
Second Exercise: AI-Assisted Development



Second Exercise: AI-Assisted Development

- Technique 1: Planning
  - After submitting a prompt in planning mode, you can view and edit the plan document
  - When ready, hit “Build”



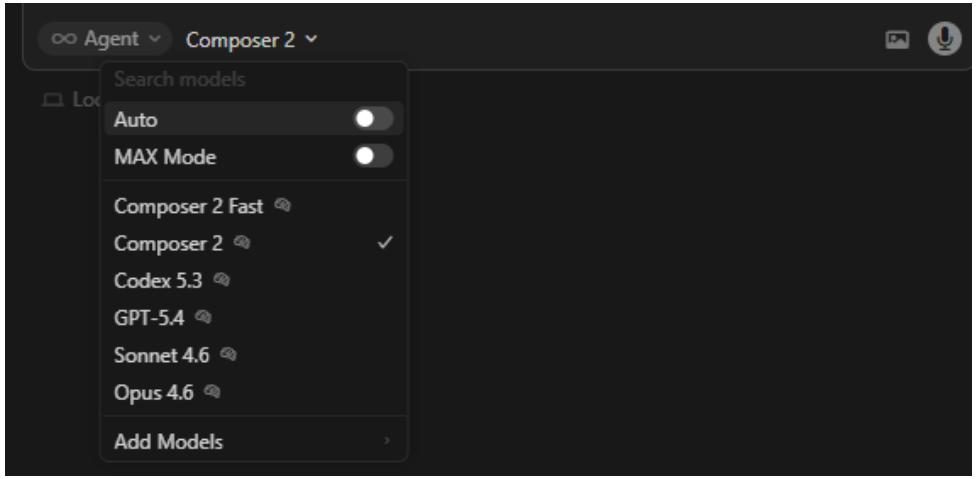


Second Exercise: AI-Assisted Development

- Technique 2: Incremental Improvement
  - Make incremental changes as small as possible, rather than trying to do the whole thing at once. The smaller and more defined the change – the more likely AI is to succeed.
  - **Too Big:** "I want the visualizer to be able to zoom in and out and move nodes around."
    - **OK:** "Add the ability to listen to keyboard input interactively."
    - **OK:** "When I press z, the visualization should zoom out by 25%."
    - **OK:** "Add the ability to listen to mouse input interactively."
    - **OK:** "When I click and drag a node, it should move (along with all the edges connected to it)"

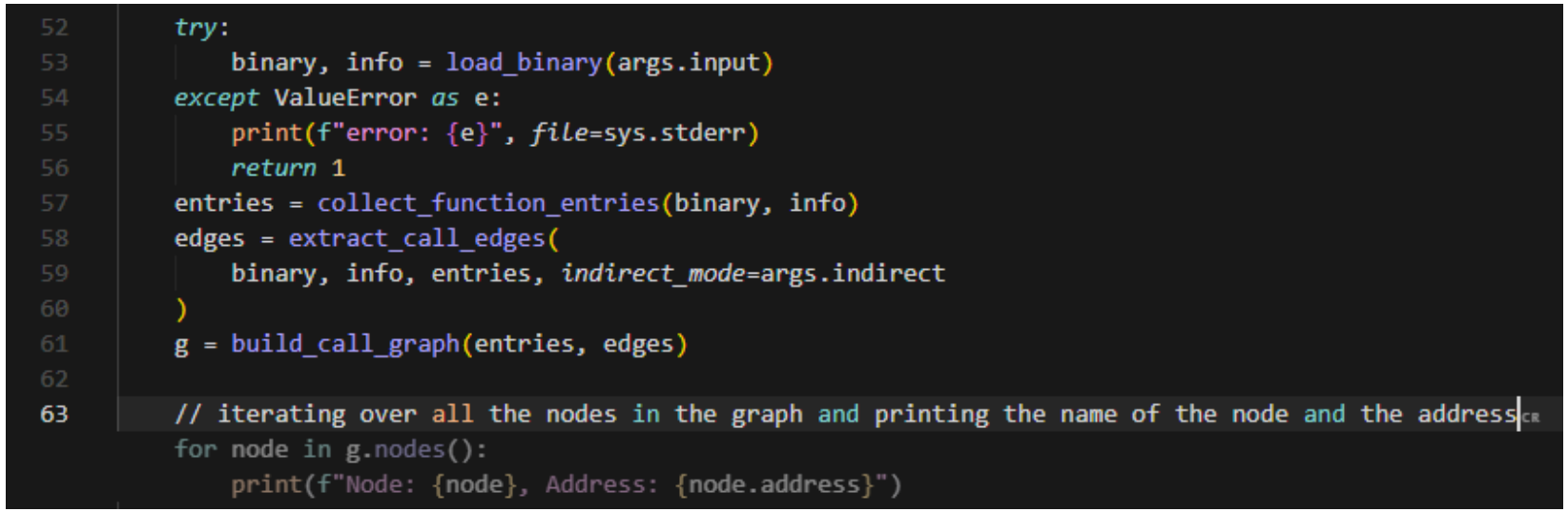
Second Exercise: AI-Assisted Development

- Technique 2: Incremental Improvement
  - You can set the specific model used here
  - The model that gets used and its level of effort is important for determining how good the results you get are
  - Small, well-specified tasks can be safely given to stupider (cheaper) models
  - The big important stuff, like coming up with the plan, can benefit greatly from using a better model



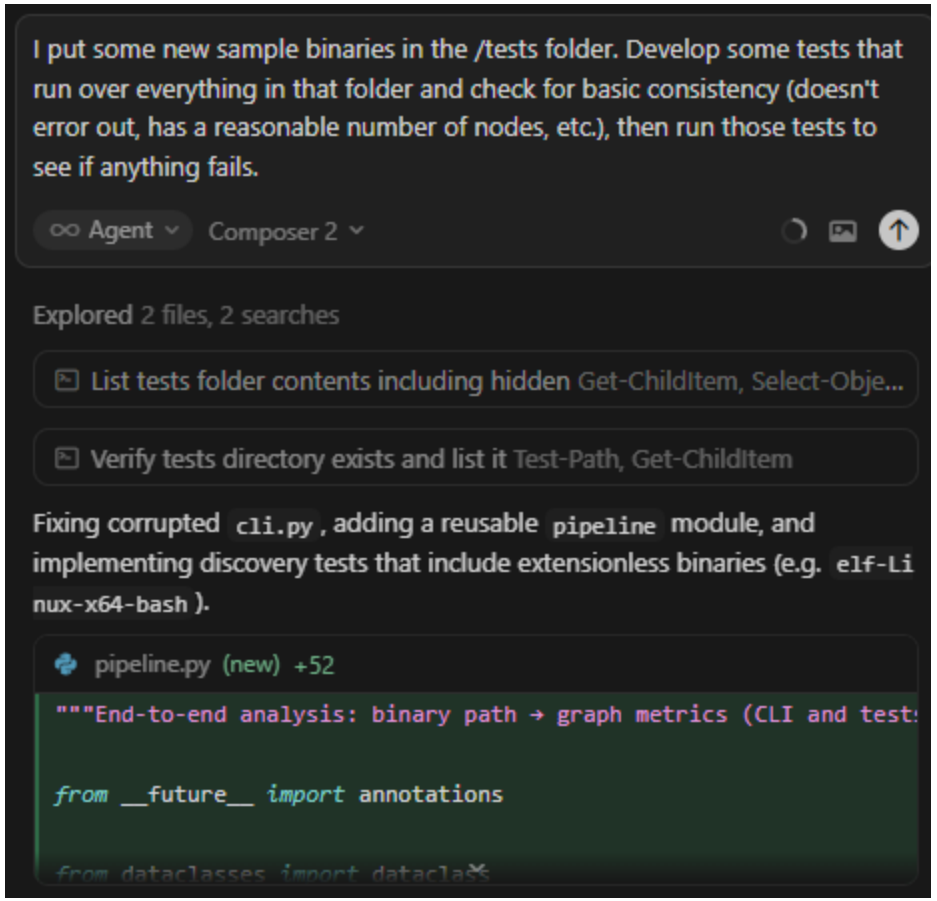
Second Exercise: AI-Assisted Development

- Technique 2: Incremental Improvement
  - Cursor has an autocomplete feature
  - Useful when you know exactly how to do something, it is a bit tricky, and you want it done right: make the surgical changes yourself
  - Try writing a comment to describe what you are about to do

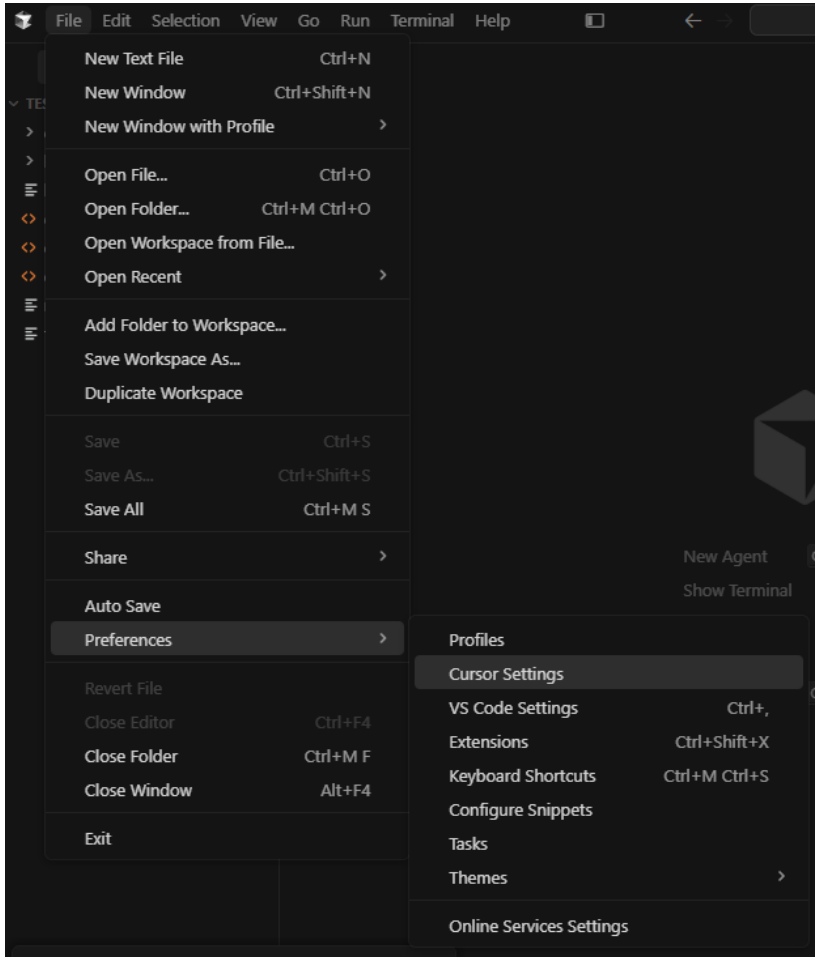


Second Exercise: AI-Assisted Development

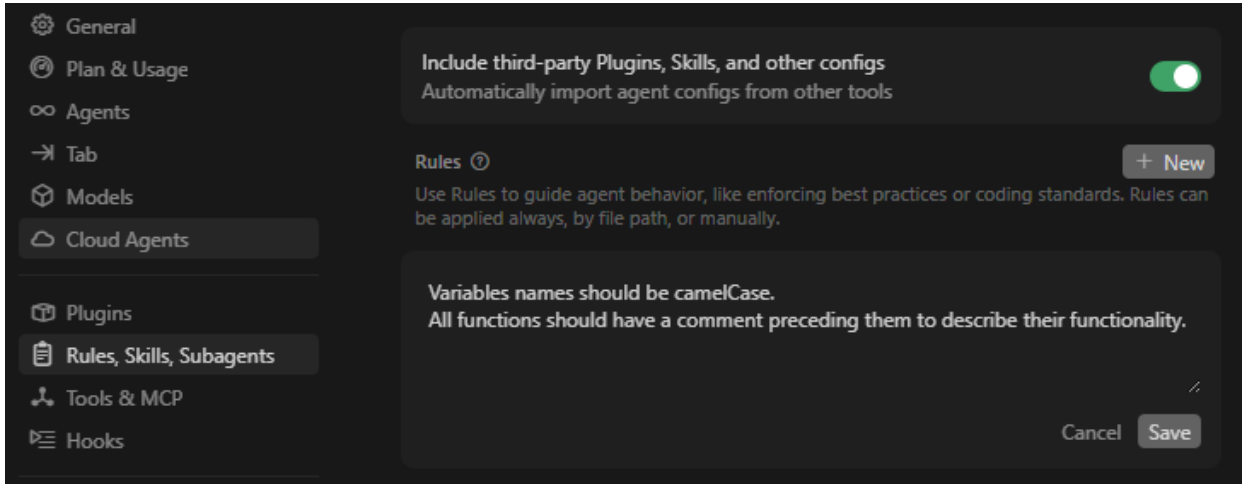
- Technique 3: Managing
  - The key to good AI coding practices is adopting many of the same good SWE practices that work for managing large software projects
  - Testing is an important one
  - If done right and well-specified, the AI can successfully debug problems in a loop on its own.
  - Be careful: in my experience, the AI decides removing tests is sometimes a good way to get things to pass the tests.



Second Exercise: AI-Assisted Development

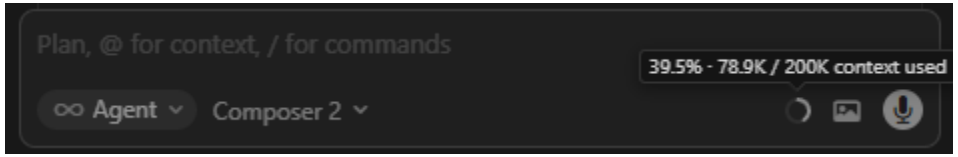


- Technique 3: Managing
  - Cursor has “rules” that allow you to more directly manage the system prompt provided to the AI. See: https://cursor.com/docs/rules
  - These rules are useful for project-level directives like formatting or coding style or overall architecture or “make sure you do/don’t do X”

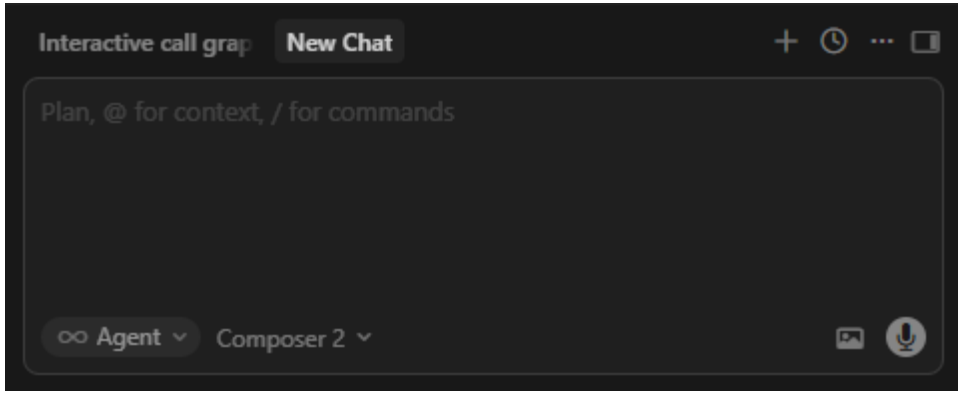


Second Exercise: AI-Assisted Development

- Technique 3: Managing
  - You can also more explicitly manage the context window of the AI.
  - See current context usage →



  - Clear context in new chat window →



  - Add specific context (file, URL, etc.) with @ →